Efficient Harmonic Balance Analysis of Waveguide Devices with Nonlinear Dielectrics

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***Abstract*—The computation of high order harmonic components of the time-harmonic field originated in waveguide structures containing nonlinear dielectrics is addressed exploiting harmonic balance and finite element methods. The solution of the nonlinear system is handled either via Picard or relaxed iterations. Being the nonlinear loop particularly time demanding, a domain decomposition method is proposed to speed-up the system solution by restricting the iteration to a small subdomain.**

***Index Terms*—Harmonic analysis, finite element methods, domain decomposition method, nonlinear media, waveguide filters.**

I. Introduction

TIME-DOMAIN analysis of waveguide devices including nonlinear dielectrics often results to be computationally prohibitive [1]. A time-harmonic finite element (FE) approach at a single frequency is not sufficient to include all relevant phenomena. Higher order harmonics are needed and may be taken into account, by using the harmonic balance finite element (HBFE) method [2], also known as multiharmonic finite element method ([3] and references therein), which provides a fast and accurate solution for low frequency problems. In this framework an algebraic domain decomposition (DD) technique was proposed to improve computational timings also in high frequency problems [4], but no harmonic balance was present there and hence only the fundamental frequency was evaluated. In this letter, [4] is expanded by implementing a full harmonic balance domain decomposition finite element (HB-DDFE) method for solving Helmholtz equation. The capability and efficiency of the method are proved on a waveguide filter example.

II. Formulation

Given a generic multiport device (Fig. 1), its domain Ω, with boundary Γ, can be divided into a large part, Ω_1, comprising only linear media, and a small part, Ω_2, comprising all nonlinear media, with $\Omega = \Omega_1 \cup \Omega_2$ and $\Omega_1 \cap \Omega_2 = \emptyset$.

The permittivity in Ω can be expressed as

$$\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) = \begin{cases} \bar{\varepsilon}_r(\mathbf{r}), & \mathbf{r} \in \Omega_1 \\ \tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) = \bar{\varepsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r} \in \Omega_2 \end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field in the generic point $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator describing the nonlinear behavior [4].

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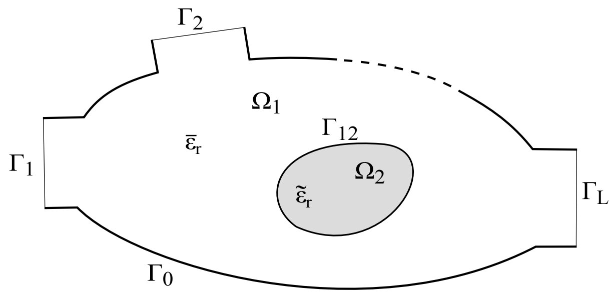

Fig. 1. Generic multiport device where the total domain Ω is subdivided into a large, linear subdomain Ω_1 and a smaller subdomain Ω_2.

The electric field satisfies, within Ω, the well known vector Helmholtz equation [5] with suitable boundary conditions on Γ. For the example shown in section III, Γ comprises a part Γ_0 which will be considered perfect electric conductor, and L parts Γ_l corresponding to the ports, where a modal expansion of the field is enforced [5]. Γ_{12} corresponds to the internal boundary between subdomains Ω_1 and Ω_2.

In a standard FE Galerkin framework the domain Ω is discretized into a set of non overlapping finite elements and the electric field is approximated as a linear combination of vector basis functions $\boldsymbol{\alpha}_n$ [5]. Then by testing Helmholtz equation over Ω with weighting functions $\boldsymbol{\alpha}_m$, chosen equal to the basis, the following weak form is obtained:

$$\int_\Omega \nabla\times\boldsymbol{\alpha}_m(\mathbf{r})\cdot\frac{1}{\mu_r}\nabla\times\mathbf{E}(\mathbf{r})\,d\Omega - k^2\int_\Omega \boldsymbol{\alpha}_m(\mathbf{r})\cdot\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) \mathbf{E}(\mathbf{r})\,d\Omega + \oint_\Gamma \boldsymbol{\alpha}_m(\mathbf{r})\cdot\hat{n}\times\frac{1}{\mu_r}\nabla\times\mathbf{E}(\mathbf{r})\,d\Gamma = 0 \quad (2)$$

being k the free space wavenumber. The multiharmonic dependence in the HBFE method is introduced by approximating the electric field as [3]

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_{p=1}^{P}\Big(\mathbf{E}_p^{(s)}(\mathbf{r})\sin(p\omega_0 t) + \mathbf{E}_p^{(c)}(\mathbf{r})\cos(p\omega_0 t)\Big) \quad (3)$$

where ω_0 is the fundamental angular frequency and, here and in the following, the hat over a quantity denotes its approximated value given by the truncated Fourier series. $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the field at $p\omega_0$ and hence they must individually satisfy Helmholtz equation with $k = pk_0$. This is done in the FE framework by using the FE expansion for each coefficient $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$. The scalar relative permittivity can also be approximated as [3]

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}, t) = \varepsilon_{r_0} + \sum_{g=1}^{G} \left(\varepsilon_{r_g}^{(s)} \sin(g\omega t) + \varepsilon_{r_g}^{(c)} \cos(g\omega t) \right) \quad (4)$$

A conventional, linear, FE approach discretization of (2) leads to a system in the form $[a_{mn}][E_n] = [b_m]$ [5]. In an HBFE approach, $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ must be substituted by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (2). By performing a further testing with weights $\sin(q\omega t)$ or $\cos(q\omega t)$, $q = 1 \dots P$, and integrating over the time period of the fundamental $\left[0, \frac{2\pi}{\omega}\right]$, a large linear system is obtained which has formally the same structure as a conventional FE system but in which each entry a_{mn}, E_n or b_m is itself a matrix or a vector:

$$a_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \dots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \dots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \dots & a_{mn}^{(PcPc)} \end{bmatrix} \quad (5)$$

$$E_n = \begin{bmatrix} E_n^{(1s)} & E_n^{(1c)} & \dots & E_n^{(Pc)} \end{bmatrix}^{\mathrm{T}} \quad (6)$$

$$b_m = \begin{bmatrix} b_m^{(1s)} & b_m^{(1c)} & \dots & b_m^{(Pc)} \end{bmatrix}^{\mathrm{T}} \quad (7)$$

The superscripts $(q[s|c])$ and $(p[s|c])$ indicates that the corresponding entry was computed by testing (2) with $[\sin(q\omega t)|\cos(q\omega t)]$ while the corresponding harmonic basis of (3) was $[\sin(p\omega t)|\cos(p\omega t)]$.

The nonlinearity is finally handled via a relaxed iteration. As a starting point a null electric field $[E]^0$ is assumed, computing the pertinent system matrix $[A_{([E]^0)}]$ and known term $[b_{([E]^0)}]$ and solving the system $[A_{([E]^0)}][E]^1 = [b_{([E]^0)}]$. The newly computed field $[E]^1$ can now be used to update system matrix and known term and perform the following relaxed iteration:

$$[A_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}][E]^i = [b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}] \quad (8)$$

with $i \geq 2$ and $\gamma \in (0, 1]$. $\gamma = 1$ gives the standard Picard iteration [4], while low γ values damps the oscillations which may arise with highly nonlinear materials or high intensity impressed fields at the cost of a slower convergence [6]. The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value τ.

The order of the resulting HBFE system is $2NP$, and this number can become very high, and, since the solutions have to be refined iteratively building at each step a new linearized system, computing time can become too high.

To enhance the computational efficiency, a substructuring DD approach is here proposed [4], [7]. The multiharmonic unknowns vector $[E]$ defined by the HBFE method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2, respectively. Unknowns belonging to the boundaries $\Gamma \cup \Gamma_{12}$ are placed in a third vector $[E^{(c)}]$. The HBFE system can then be recast in :

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M^{(2)}]^{i-1} & [P^{(2)}]^{i-1} \\ [Q^{(1)}] & [Q^{(2)}]^{i-1} & [C]^{i-1} \end{bmatrix} \begin{bmatrix} [E^{(1)}]^i \\ [E^{(2)}]^i \\ [E^{(c)}]^i \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b^{(2)}]^{i-1} \\ [b^{(c)}]^{i-1} \end{bmatrix} \quad (9)$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$, with null harmonic coupling coefficients, while $[M^{(2)}]$ contains coefficients for the nonlinear subdomain Ω_2. $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between interior unknowns and boundary unknowns collected in $[E^{(c)}]$.

By using the Schur complement concept, the boundary unknowns $[E^{(c)}]^i$ of (9), and hence the generalized scattering matrix of the device, can be retrieved [4], [7].

Hence, in order to solve the HBDDFE system, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 and Γ_{12} are to be updated. The efficiency of this DD technique relies on the fragmentation of the matrices and the subsequent computation of partial solutions. When the number of coefficients related to the nonlinear media is small, noticeable improvements can be achieved.

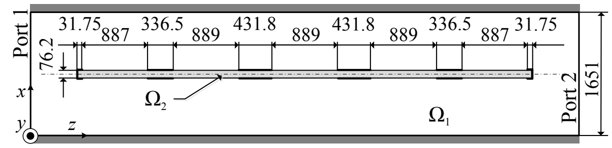

Fig. 2. Cross section (H-plane) of a passband filter realized by placing on the E-plane a dielectric slab partially metalized on both sides (slab is shown in light grey; metal strips - uniform along y are shown as thick black lines) in a WR6 rectangular waveguide. Measures are given in μm.

III. Numerical Results

A millimeter wave passband filter in WR6 is here analyzed (Fig. 2) [4]. Due to symmetries, the problem can be treated as 2D on the H-plane. First order nodal elements are used and the discretization leads to 2470 degrees of freedom, 42 of which belong to the nonlinear subdomain Ω_2, 456 to the boundaries and 1972 to the linear subdomain Ω_1. Such a fine discretization is necessary to obtain a good approximation of the field up to the 5^{th} harmonic.

All conductors are considered perfect. The continuity of the field at the ports is imposed through a modal expansion exploiting only TE_{m0} modes, $m = 2n+1$, $n = 1, \dots, 4$. The excitation is a TE_{10} mode impinging at Port 1 with maximum field amplitude E_i.

The dielectric, enclosed in Ω_2, presents a Kerr-type nonlinearity of such that [4], [8]

$$\tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\varepsilon}_r(\mathbf{r}) + \alpha_2 \, |\mathbf{E}(\mathbf{r})|^2, \qquad \mathbf{r} \in \Omega_2 \quad (10)$$

where $\bar{\varepsilon}_r(\mathbf{r}) = 2.1$ and $\alpha_2 = 1.625 \ 10^{-10} \ \mathrm{m}^2/\mathrm{V}^2$ [9]. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus even ones can been neglected. Furthermore, as a sinusoidal excitation is chosen, only $\sin(p\omega t)$

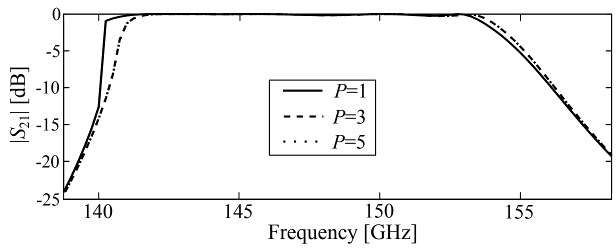

Fig. 3. Behavior of the TE_{10} spectral response as the number of harmonics retained for the HBFE increases ($E_i = 10$kV/m).

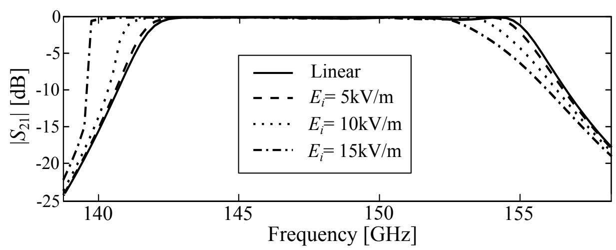

Fig. 4. $|S_{21}|$ of the nonlinear filter for several values of E_i compared to the linear case. Field is approximated up to the 5^{th} harmonic.

related coefficients are retained. For the relaxed iteration stop criterion, $\tau = 10^{-6}$ has been chosen.

As a first analysis, the effect of the total number of harmonics P retained on the filter's spectral response is investigated (Fig. 3). For an incident field amplitude $E_i = 10$kV/m (corresponding to an incident power equal to 72mW) the Picard iteration proved to converge. It is evident from Fig. 3 that including the 3^{rd} harmonic is crucial for proper evaluation of the device bandwidth.

To gain better insight, simulations have been repeated with $E_i = 5$kV/m and $E_i = 15$kV/m (18mW and 162mW of incident power, respectively), for $P = 5$. The results, compared to those of a linear device ($\alpha_2 = 0$), are shown in Fig. 4. There is an evident shift of the band towards lower frequencies as higher power densities are involved. Furthermore the $E_i = 15$kV/m did not converge with Picard iteration and required a relaxed $\gamma = 0.1$ iteration. In this case, the abrupt variation of $|S_{21}|$ around 139.5GHz points out that $P = 5$ is not sufficient to accurately analyze the device. Indeed, the power associated to harmonics grows as the incident field increases as shown in Fig. 5.

The efficiency of the DD technique can be assessed by its acceleration, that is the ratio between time for a full domain computation - inclusive of nonlinear iterations - and the corresponding DD computation [4]. Fig. 6 presents the acceleration for each frequency point, and for different values of P at $E_i = 10$kV/m.

In general, higher order HBDDFE systems have lower acceleration, since the matrices dimensions related to nonlinear media increase. With a $P = 3$ system, the acceleration varies in the range [2.51, 9.15], averaging to 4.18, and leading to an HBDDFE solution in 24% of the time required by a full domain HBFE solution.

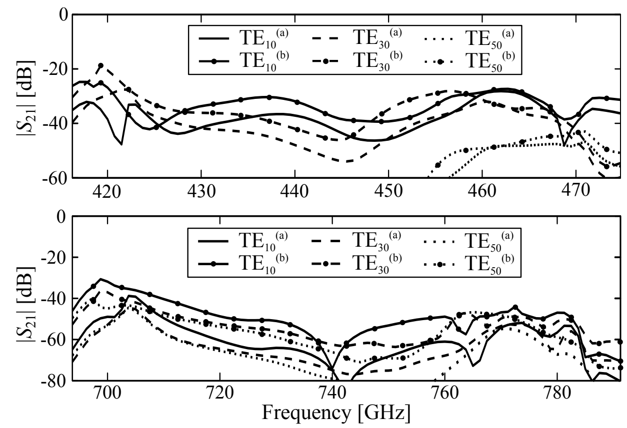

Fig. 5. Spectral responses at 3^{rd} (top) and 5^{th} (bottom) harmonics for scattered modes (Incident TE_{10} with (a) $E_i = 10$kV/m; (b) $E_i = 15$kV/m.

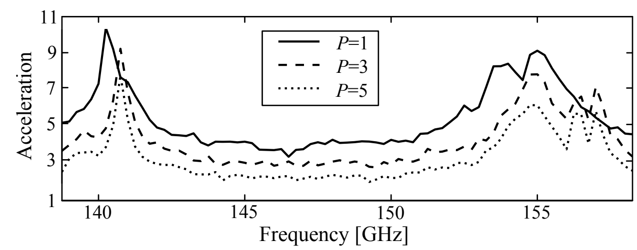

Fig. 6. Acceleration spectrum obtained for an HBFE system of several harmonic orders. The computations are made for $E_i = 10$kV/m.

IV. Conclusion

A harmonic balance approach to the DD-FE solution for the efficient analysis of waveguide devices containing nonlinear dielectrics has been presented and the accuracy of the multiharmonic solution discussed.

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